

Compact and Finite-Rank Operators (Quick Notes)

Setting

Let X, Y be normed (often Banach) spaces over $\mathbb{R}$ or $\mathbb{C}$, and let $\mathcal{B}(X, Y)$ denote the space of bounded linear operators $T : X \rightarrow Y$ with the operator norm

$$\|T\| = \sup_{\|x\| \leq 1} \|Tx\|.$$

1 Finite-Rank Operators

Definition 1 (Finite rank). *A bounded linear operator $T : X \rightarrow Y$ has finite rank if its range $T(X)$ is a finite-dimensional subspace of Y . The rank of T is $\dim T(X)$.*

Example 1.

- **Rank one.** *If $f \in X'$ and $y_0 \in Y$, the map $T(x) = f(x)y_0$ has range $\text{span}\{y_0\}$, hence rank 1.*
- **Finite sum of rank ones.** *If $T(x) = \sum_{k=1}^n f_k(x)y_k$ with $f_k \in X'$ and $y_k \in Y$, then $T(X) \subseteq \text{span}\{y_1, \dots, y_n\}$, so T has finite rank.*
- **Coordinate truncation on ℓ^2 .** *$T(x_1, x_2, \dots) = (x_1, \dots, x_n, 0, 0, \dots)$ has rank n .*

2 Compact Operators

Definition 2 (Compact operator). *A bounded linear operator $T : X \rightarrow Y$ is compact if it maps bounded sets into relatively compact sets. Equivalently: for every bounded sequence $(x_m) \subset X$ there is a subsequence (x_{m_j}) such that (Tx_{m_j}) converges in Y .*

Example 2.

- **Finite rank $\Rightarrow$ compact.** *In finite-dimensional spaces, bounded sets are relatively compact (Heine–Borel). Thus every finite-rank operator is compact.*
- **Diagonal operator on ℓ^2 .** *Define $T : \ell^2 \rightarrow \ell^2$ by*

$$T(x_1, x_2, \dots) = \left(\frac{x_1}{1}, \frac{x_2}{2}, \frac{x_3}{3}, \dots\right).$$

Then T is compact: the singular values $1/n \rightarrow 0$, so T is the norm-limit of finite-rank truncations.

- **Integral operator on $C([0, 1])$.** *If $K \in C([0, 1]^2)$ and*

$$(Tf)(x) = \int_0^1 K(x, y)f(y) dy,$$

then $T : C([0, 1]) \rightarrow C([0, 1])$ is compact (Arzelà–Ascoli).

Remark 1 (Non-examples).

- The identity I_X on an infinite-dimensional Banach space X is not compact: the unit ball is bounded but not relatively compact.
- The inclusion $\iota : \ell^2 \hookrightarrow \ell^\infty$ is bounded but not compact: (e_n) lies in the ℓ^2 unit ball and (ιe_n) has no convergent subsequence in ℓ^∞ .

3 Basic Theorems

Proposition 1. *Every finite-rank operator is compact.*

Proof sketch. If $\dim T(X) < \infty$, then $T(B)$ is bounded in a finite-dimensional space, hence relatively compact. Apply the definition. $\square$

Theorem 1 (Closedness and ideal properties). *Let X, Y, Z be Banach spaces. The set $\mathcal{K}(X, Y)$ of compact operators is a closed linear subspace of $\mathcal{B}(X, Y)$. Moreover, if $A \in \mathcal{B}(Y, Z)$ and $B \in \mathcal{B}(X, Y)$, then*

$$A \circ T \circ B \in \mathcal{K}(X, Z) \quad \text{whenever } T \in \mathcal{K}(Y).$$

Thus $\mathcal{K}$ is a two-sided operator ideal.

Proposition 2 (Riesz lemma consequence). *If X is infinite-dimensional and $T \in \mathcal{B}(X)$ is compact and injective, then T cannot have a bounded inverse on $T(X)$; in particular, the identity operator is not compact.*

Theorem 2 (Sequential characterization). *$T : X \rightarrow Y$ is compact $\iff$ for every bounded sequence $(x_m) \subset X$ there exists a subsequence (x_{m_j}) such that (Tx_{m_j}) converges in Y .*

Proposition 3 (Weak-to-strong property). *If X is a Banach space and $T \in \mathcal{K}(X, Y)$, then whenever $x_n \rightharpoonup 0$ weakly in X and (x_n) is bounded, we have $\|Tx_n\| \rightarrow 0$.*

4 Relations and Remarks

- Finite-rank $\subset$ compact $\subset$ bounded, with both inclusions proper in infinite dimensions.
- Compact operators generalize “finite-matrix” behavior: spectral theory for compact operators on infinite-dimensional Hilbert spaces mimics the finite-dimensional case (e.g. nonzero spectrum is discrete with 0 as the only possible accumulation point).
- In many classical spaces (e.g. Hilbert spaces, $C(K)$ with K compact metric), compact operators can be approximated in norm by finite-rank operators; in full generality this relies on approximation properties of the spaces involved.

Mini Checklist / Examples at a Glance

- Rank-one: $T(x) = f(x)y_0$ (finite-rank $\Rightarrow$ compact).
- Diagonal on ℓ^2 with entries $s_n \rightarrow 0$ is compact; if $s_n \not\rightarrow 0$, it is not compact.
- Integral operator with continuous kernel on $C([0, 1])$: compact (Arzelà–Ascoli).
- Identity on infinite-dimensional space: not compact.